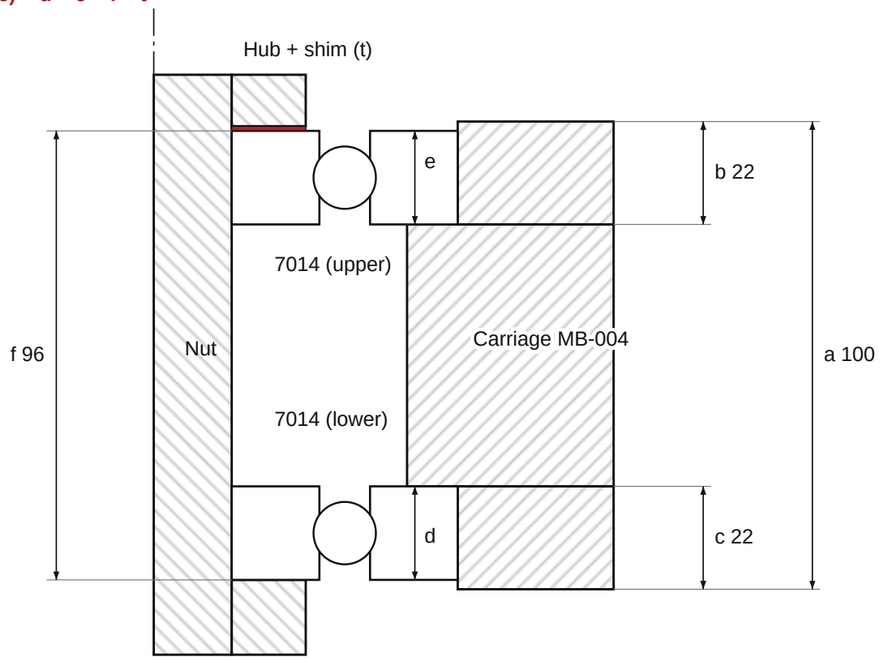
STACK A: BEARING PRELOAD, AXIAL (worst case, RSS, Monte Carlo)

δ is the axial squeeze on the 7014 O-pair when the hub is tightened: + is preload, − is axial play. Required window: **+0.005 to +0.035 mm** (light preload *).

δ = (a − b − c) + d + e − f + t



Ref	Link	Source	Sign	Nominal	Tolerance	Mean	± half
a	Carriage height	MB-004, ISO 2768-m	+	100	±0.30	100.000	0.300
b	Upper bore depth	MB-004, ISO 2768-m	−	22	±0.20	22.000	0.200
c	Lower bore depth	MB-004, ISO 2768-m	−	22	±0.20	22.000	0.200
d	Bearing width, lower	ISO 492 Normal, 7014	+	20	0 / −0.15	19.925	0.075
e	Bearing width, upper	ISO 492 Normal, 7014	+	20	0 / −0.15	19.925	0.075
f	Nut clamp length	MB-005-02, ISO 2768-m *	−	96	±0.30	96.000	0.300

Results, δ in mm

Case	Mean δ	Worst case	RSS	In window
1. As drawn	−0.150	−1.300 to +1.000	−0.671 to +0.371	4.3 %
2. Tightened: web 56 ±0.02, f ±0.02	+0.020	−0.170 to +0.210	−0.090 to +0.130	31.8 %
3. Measured shim (adopted)	+0.020	+0.005 to +0.035	+0.011 to +0.029	100 %

Monte Carlo: 2,000,000 assemblies, each link normal with 3σ = tolerance, truncated at the limit. "In window" is the share inside +0.005 to +0.035.

FINDINGS AND ACTIONS, STACK A

1. **General tolerances cannot set preload.** As drawn, the worst case spans 2.3 mm against a 0.03 mm window, and only 4.3 % of assemblies land inside it.
2. **Tightening does not rescue it.** With the machined links at ±0.02, the bearing width tolerance alone (±0.15 for the pair) is still ten times the window. Only 31.8 % pass, at higher machining cost.
3. **Adopted: measure, then shim.** Keep MB-004 as drawn. After anodizing, measure the assembled gap, grind one shim between hub and upper inner ring. Links: measurement ±0.005, shim grinding ±0.005, 0.01 mm step rounding ±0.005. Result ±0.015 about a +0.020 target.
4. **Applied on MB-005-02 Rev A:** nominal f is 97.5, so the shim is always positive. Shim range 0.52 to 2.82 mm worst case; t = 0.020 − (a − b − c + d + e − f).

STACK B: SCREW CLEARANCE AT BOTTOM STEADY, RADIAL

Does the carriage push the screw into the steady bush (item 16)? Limit: **1.000 mm** radial clearance, MB-005 note 2.

Ref	Link (radial, one direction)	Source	± half
g	Rail parallel to screw, zone 0.1	MB-005 note 2	0.0500
h	Shaft Ø25 g6 in bushing, max. diametral clearance 0.120	g6 drawn; bore Ø25 +0.04/+0.10 *	0.0600
i	Ø40 H7 position Ø0.05 A B	MB-004	0.0250
j	Ø110 H7 coaxiality Ø0.015 A	MB-004	0.0075
k	Bearing OD in Ø110 H7, clearance 0 to 0.050	MB-004, ISO 492	0.0250
m	Outer ring radial run-out Kea 0.035	ISO 492 Normal	0.0175
n	Inner ring radial run-out Kia 0.020	ISO 492 Normal	0.0100
p	Bottom steady location to rails	Allocated *	0.2500
	Worst case / RSS offset		**0.445 / 0.265**

Pass. Margin 0.555 mm worst case, 0.735 mm RSS. Link p can open up to ±0.805 mm with no other change, so the steady bracket needs no precision machining.

Notes

1. Units mm. Links marked * are assumptions, to be confirmed: preload window against the bearing maker's data, bushing bore against the catalogue. Case 1 uses the initial nut length f = 96 with ISO 2768-m; MB-005-02 Rev A now carries f = 97.5.
2. Hub assumed to seat on a hard stop on the nut. Shoulder perpendicularity 0.01 A affects bearing tilt, not length, and is excluded.
3. Differential expansion (aluminium 23.8, steel 12, bronze 18 × 10^{-6}/K *) adds about +0.003 mm at +30 K; not in the budget above.
4. Design study: nothing has been built or measured. MB-004 Rev B stays valid; the shim action is applied on MB-005-02.

Material: -	Standards: ISO 2768, ISO 492	Method: WC, RSS, Monte Carlo	Mass: -
Scale: NTS	Tolerancing principle: ISO 8015	Designed by Mahdi Bachraoui, 2026-11-08	Checked by: -
Document type: Calculation	Tolerance stack-up Vertical axis MB-005, carriage MB-004 Rev B		Format: A3
Drawing Number: MB-007			Rev A Sheet 1/1